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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Database Systems (course)

Course outline

[About NPTEL \(\)](#)[Introduction to Database Systems \(\)](#)[Week 0 \(\)](#)[Week 1 \(\)](#)[Week 2 \(\)](#)[Week 3 \(\)](#)

Week 5 : Assignment 5

The due date for submitting this assignment has passed.

Due on 2026-02-25, 23:59 IST.

Assignment submitted on 2026-02-18, 09:23 IST

Use the following schema of the academic institution relational database in the following questions wherever required.

student(rollNo , name, degree, year, sex, deptNo, advisor)

department(deptId , name, hod, phone)

professor(empld , name, sex, startYear, deptNo, phone)

course(courseld , cname, credits, deptNo)

enrollment(rollNo, courseld, sem, year , grade)

teaching(empld, courseld, sem, year , classRoom)

Week 4 ()**Week 5 ()**

- ☐ Data definition using SQL (unit? unit=47&lesson=48)
- ☐ Basic SQL query block and subqueries (unit? unit=47&lesson=49)
- ☐ Correlated subqueries (unit? unit=47&lesson=50)
- ☐ Lecture Slides (unit? unit=47&lesson=51)
- ☐ Week 5 Feedback Form: Introduction to Database Systems!! (unit? unit=47&lesson=52)

**● Quiz: Week 5 :
Assignment 5
(assessment?
name=194)**

Week 6 ()**Week 7 ()****Week 8 ()****Week 9 ()**

preRequisite(preCourseId, courseId)

deptNo is a foreign key in the *student* , *professor* and *course* relations referring to *deptId* of *department* relation;

advisor is a foreign key in the *student* relation referring to *empId* of *professor* relation;

hod is a foreign key in the *department* relation referring to *empId* of *professor* relation;

rollNo is a foreign key in the *enrollment* relation referring to *rollNo* of *student* relation;

courseId is a foreign key in the *enrollment* , teaching relations referring to *courseId* of *course* relation;

empId is a foreign key of the teaching relation referring to *empId* of *professor* relation;

preCourseId and *courseId* are foreign keys in the *preRequisite* relation referring to *courseId* of the *course* relation.

1) Which of the following is not a Data Definition Language keyword of SQL?

- ☐ create
- ☒ insert
- ☐ alter
- ☐ drop

Yes, the answer is correct.

Score: 2

Accepted Answers:

insert

2 points

[Week 10 \(\)](#)[Week 11 \(\)](#)[Week 12 \(\)](#)[Download Videos \(\)](#)[Lecture Notes \(\)](#)[Problem Solving
Session - Jan 2026 \(\)](#)

2) What is the relation between NOT NULL and DEFAULT keywords in the attributes' specification of the relation definitions?

2 points

- ☒ Both can be specified independently
- ☐ NOT NULL keyword must be specified only if DEFAULT keyword is specified
- ☐ If NOT NULL keyword is specified then DEFAULT keyword must be specified
- ☐ NOT NULL keyword must be specified if and only if DEFAULT keyword is specified

Yes, the answer is correct.

Score: 2

Accepted Answers:

Both can be specified independently

3) Which of the following key constraints can be specified only once in a relation definition?

2 points

(i) primary key (ii) foreign key (iii) unique key

- ☐ Only (iii)
- ☒ Only (i)
- ☐ Only (i) and (iii)
- ☐ All (i), (ii) and (iii)

Yes, the answer is correct.

Score: 2

Accepted Answers:

Only (i)

4) Which of the following is the negation of =ALL operator of SQL?

2 points

- ☐ <> ALL
- ☐ NOT ALL
- ☒ <> ANY
- ☐ NOT EXISTS

Yes, the answer is correct.

Score: 2

Accepted Answers:

$\neq$ ANY

5) Consider the following query to retrieve the course(s) with highest number of credits :

2 points

```
SELECT c1.courseld
FROM course c1
WHERE c1.credits > ALL (SELECT c2.credits
                        FROM course c2
                        WHERE -----)
```

Which of the following options is correct regarding the WHERE clause of the sub-query ?

- ☒ The condition should be: $c2.courseld \neq c1.courseld$
- ☐ The WHERE clause isn't required
- ☐ The condition should be: $c2.credits > c1.credits$
- ☐ It can not be completed to solve the given query

No, the answer is incorrect.

Score: 0

Accepted Answers:

The condition should be: $c2.credits > c1.credits$

6) Which of the following queries would find the courses with at least two prerequisites?

3 points

- ☐

```
SELECT *
FROM course c1
WHERE EXISTS (SELECT * FROM prerequisite p1
              WHERE p1.courseld = c1.courseld)
AND EXISTS (SELECT * FROM prerequisite p2
            WHERE p2.courseld = c1.courseld)
```
- ☐

```
SELECT *
FROM course c1
```

WHERE EXISTS (SELECT * FROM prerequisite p1
 WHERE p1.courseld = c1.courseld)
 AND EXISTS (SELECT * FROM prerequisite p2
 WHERE p2.courseld = c1.courseld
 AND p1.preCourseld <> p2.preCourseld)

☐

SELECT *
 FROM course c1
 WHERE EXISTS (SELECT * FROM prerequisite p1, prerequisite p2
 WHERE p1.courseld = c1.courseld
 AND p2.courseld = c1. courseld)

☒

SELECT *
 FROM course c1
 WHERE EXISTS (SELECT * FROM prerequisite p1, prerequisite p2
 WHERE p1.courseld = c1.courseld
 AND p2.courseld = c1. courseld
 AND p1.preCourseld <> p2.preCourseld)

Yes, the answer is correct.

Score: 3

Accepted Answers:

*SELECT *
 FROM course c1
 WHERE EXISTS (SELECT * FROM prerequisite p1, prerequisite p2
 WHERE p1.courseld = c1.courseld
 AND p2.courseld = c1. courseld
 AND p1.preCourseld <> p2.preCourseld)*

7) Which of the following queries would find the name of the students who got 'S' grade in all the prerequisites of course with id 200? **3 points**

☐

SELECT s.name
 FROM student s
 WHERE NOT EXISTS (SELECT p1. preCourseld
 FROM prerequisite p1

```
WHERE p1.courseld=200
AND NOT EXISTS (SELECT *
                 FROM enrollment e2
                 WHERE e2.courseld = p1.preCourseld
                 AND e2.grade <> 'S'
                 AND e.rollNo = s.rollNo))
```



```
SELECT s.name
FROM student s, enrollment e
WHERE s.rollNo = e.rollNo AND e.grade = 'S'
AND e.courseld = ALL ( SELECT p1.preCourseld
                      FROM prerequisite p1
                      WHERE p1.courseld=200)
```



```
SELECT s.name
FROM student s
WHERE NOT EXISTS ( SELECT p1. preCourseld
                  FROM prerequisite p1
                  WHERE p1.courseld=200
                  AND NOT EXISTS (SELECT *
                                   FROM enrollment e2
                                   WHERE e2.courseld = p1.preCourseld
                                   AND e2.grade = 'S'
                                   AND e.rollNo = s.rollNo))
```



```
SELECT s.name
FROM student s
WHERE EXISTS (SELECT p1. preCourseld
              FROM prerequisite p1
              WHERE p1.courseld=200
              AND NOT EXISTS (SELECT *
                               FROM enrollment e2
                               WHERE e2.courseld = p1.preCourseld
                               AND e2.grade = 'S'
                               AND e.rollNo = s.rollNo))
```

Yes, the answer is correct.

Score: 3

Accepted Answers:

```
SELECT s.name
FROM student s
WHERE NOT EXISTS ( SELECT p1. preCourseId
                   FROM prerequisite p1
                   WHERE p1.courseId=200
                   AND NOT EXISTS (SELECT *
                                   FROM enrollment e2
                                   WHERE e2.courseId = p1.preCourseId
                                   AND e2.grade = 'S'
                                   AND e.rollNo = s.rollNo))
```

8) Consider the following query

2 points

```
SELECT p1.empld, p2.empld
FROM professor p1, professor p2
WHERE p1.startYear = p2.startYear
```

Assuming that startYear is not null for any row, which one of the following statements strongly describes the result of the above query?

- ☒ It includes all the pairs of professors with same number of years of experience
- ☐ It doesn't include any of the pairs of professors with same number of years of experience
- ☐ It includes some of the pairs of professors with same number of years of experience
- ☐ It doesn't include some of the pairs of professors with same number of years of experience

Yes, the answer is correct.

Score: 2

Accepted Answers:

It includes all the pairs of professors with same number of years of experience

9) Consider the following query:

2 points

```
SELECT p1.empld, p2.empld
```

```
FROM professor p1, professor p2  
WHERE p1.startYear = p2.startYear
```

Consider an instance of professor relation with 10 rows in which startYear is not null and is distinct for all the rows. The size of the result of the query is:

- ☐ 0
- ☒ 10
- ☐ 90
- ☐ 100

Yes, the answer is correct.

Score: 2

Accepted Answers:

10